

## Week 3 Introduction

### Mathematics for Multidimensional Data

✓ **Video:** 3.1 Multidimensional Data Points and Features  
2 min

✓ **Reading:** Multidimensional Data Points and Features Recap  
10 min

📖 **Practice Quiz:** Multidimensional Data Points and Features – Review Information  
1 question

▶ **Video:** 3.2 Multidimensional Mean  
2 min

📖 **Reading:** Multidimensional Mean Recap  
10 min

📖 **Practice Quiz:** Multidimensional Mean – Review Information  
1 question

▶ **Video:** 3.3 Dispersion: Multidimensional Variables  
3 min

📖 **Reading:** Multidimensional Variables Recap  
10 min

📖 **Practice Quiz:** Dispersion: Multidimensional Variables – Review Information  
3 questions

▶ **Video:** 3.4 Distance Metrics  
5 min

📖 **Reading:** Distance Metrics Recap  
10 min

📖 **Practice Quiz:** Distance Metrics – Review Information  
4 questions



# Multidimensional Data Points and Features Recap

A few quick takeaways:

A **matrix** is a rectangular array of numbers, a way of packing numbers, or arrays of numbers (vectors) together, like so:

$$\mathbf{X} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{pmatrix}$$

A **vector** can be seen as a matrix with either one row, or one column:

$$\mathbf{Y} = \begin{pmatrix} 4 \\ 6 \\ 2 \\ 4 \\ 2 \end{pmatrix},$$

$$\mathbf{Z} = (4 \quad 6 \quad 2 \quad 4 \quad 2).$$

In the example presented at the end of the previous video, we had the marks of our students packed in one matrix, the first column being the English score, the second the Maths score:

$$\mathbf{X} = (X_1, X_2) = \begin{pmatrix} 65 & 55 \\ 70 & 52 \\ 45 & 42 \\ 61 & 34 \\ 37 & 31 \end{pmatrix}$$

We can now read the matrix either horizontally or line by line, that is, student by student, or we can do so vertically, looking at the English (the vector  $X_1$ ) or Maths scores (the vector  $X_2$ ).

$$\text{The general matrix is: } \mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} =$$

$(X_1, X_2, \dots, X_p)$ , where we would have  $p$  different scores per student, so  $p$  score vectors. In an abstract case mathematicians prefer to call  $p$  using the letter  $j$ , so we will switch to that now. Since we have  $n$  students (observations) per subject (variable), our vector corresponding to each